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[ ]

```
import cv2
```

[5]
13s

```
# Downloading required libraries

!pip install opencv-python
!pip install opencv-contrib-python
!pip install matplotlib
!pip install numpy
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12
Requirement already satisfied: numpy<2.3.0,>=2 in /usr/local/lib/python3.
Requirement already satisfied: opencv-contrib-python in /usr/local/lib/py
Requirement already satisfied: numpy<2.3.0,>=2 in /usr/local/lib/python3.
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/di
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.12/d
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dis
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/pyt
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-pa
```

[ ]

```
# ============================================
# Install and import required libraries
# ============================================

# OpenCV already exists in Colab, but we import it
import cv2
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files
```

[ ]

```
from google.colab import files

# ============================================
# Upload MRI image
# ============================================

print(">>> Please upload your MRI image file (.jpg / .jpeg / .png)")

uploaded = files.upload()
```

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[10] 0s

```
import cv2
```

[11] 0s

```
# ============================================
# Install and import required libraries
# ============================================

# OpenCV already exists in Colab, but we import it
import cv2
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files
```

[12] 56s

```
from google.colab import files

# ============================================
# Upload MRI image
# ============================================

print(">>> Please upload your MRI image file (.jpg / .jpeg / .png)")

uploaded = files.upload()

# Getting file name
for name in uploaded.keys():
    filename = name

    print("Image uploaded successfully:", filename)
```

... Show hidden output

[13] 0s

```
# ============================================
# Reading image using OpenCV
# ============================================

img = cv2.imread(filename)

if img is None:
    raise ValueError("Image not loaded! Check file format or upload aga

    # Convert BGR to RGB for correct display
    rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    plt.imshow(rgb)
    plt.title("Original MRI Image")
    plt.axis("off")
    plt.show()
```

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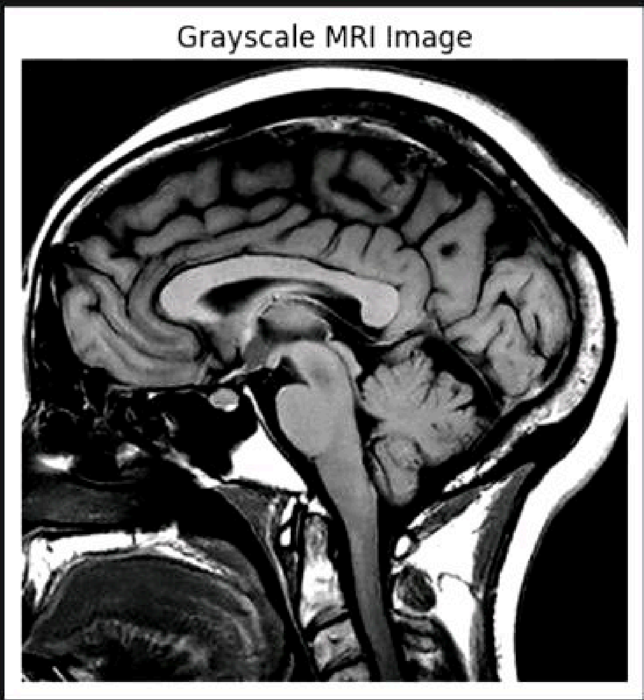
[14]

```
# ==============================================
# Converting to grayscale
# ==============================================

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.imshow(gray, cmap='gray')
plt.title("Grayscale MRI Image")
plt.axis("off")
plt.show()
```

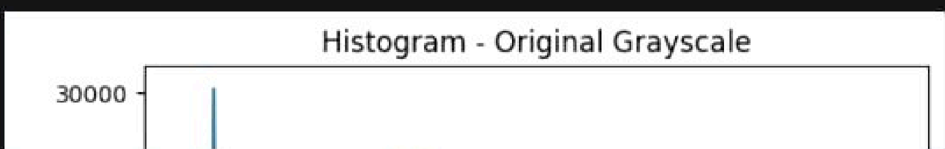


[15]

```
# ==============================================
# Calculating histogram of original grayscale image
# ==============================================

hist_original = cv2.calcHist([gray], [0], None, [256], [0,256])

plt.figure(figsize=(6,4))
plt.plot(hist_original)
plt.title("Histogram - Original Grayscale")
plt.xlabel("Intensity")
plt.ylabel("Pixel Count")
plt.show()
```



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[15]

```
# ==========================================
# Calculating histogram of original grayscale image
# ==========================================

hist_original = cv2.calcHist([gray], [0], None, [256], [0,256])

plt.figure(figsize=(6,4))
plt.plot(hist_original)
plt.title("Histogram - Original Grayscale")
plt.xlabel("Intensity")
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plt.show()
```
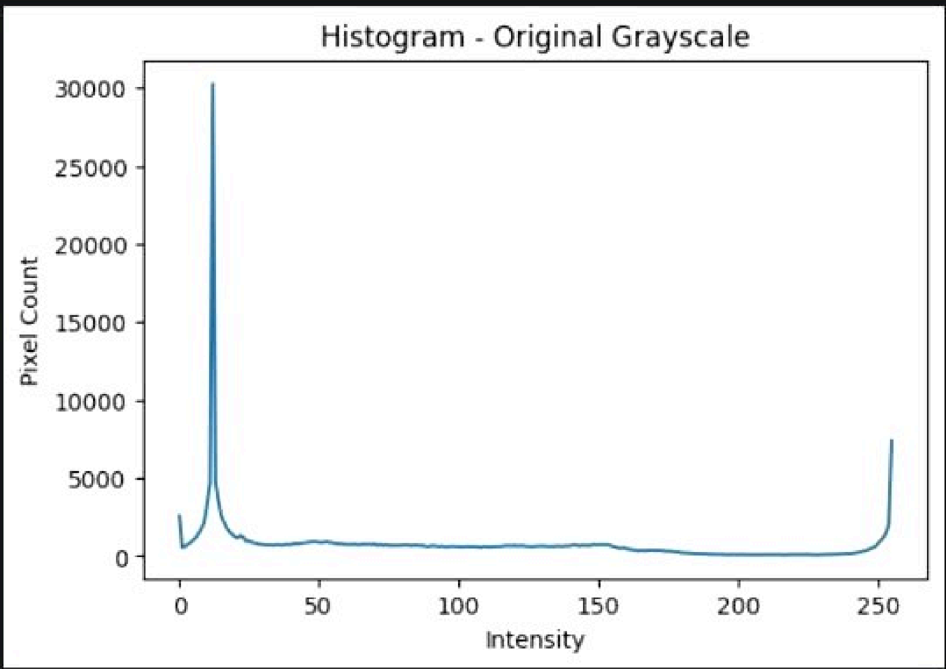
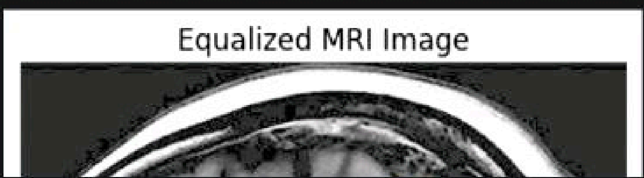



[16]

```
# ==========================================
# Applying Histogram Equalization
# ==========================================

equalized = cv2.equalizeHist(gray)

plt.imshow(equalized, cmap='gray')
plt.title("Equalized MRI Image")
plt.axis("off")
plt.show()
```



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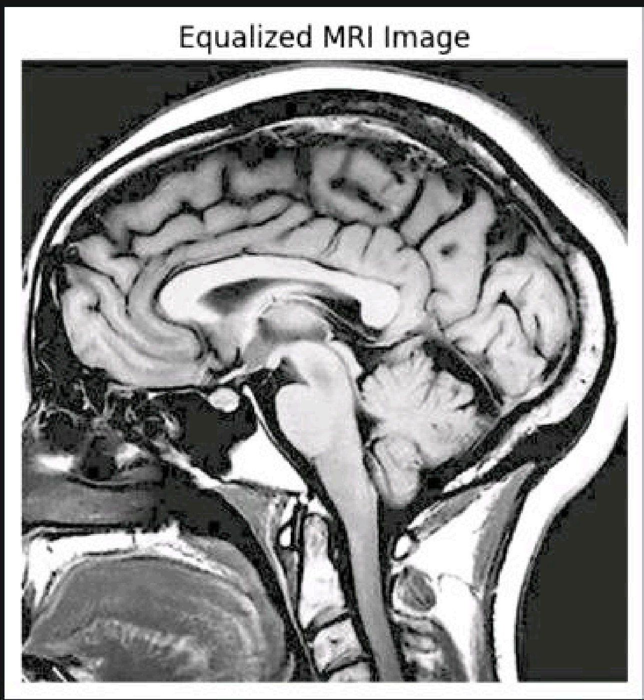
[16] 1s

```
# ============================================
# Applying Histogram Equalization
# ============================================

equalized = cv2.equalizeHist(gray)

plt.imshow(equalized, cmap='gray')
plt.title("Equalized MRI Image")
plt.axis("off")
plt.show()
```

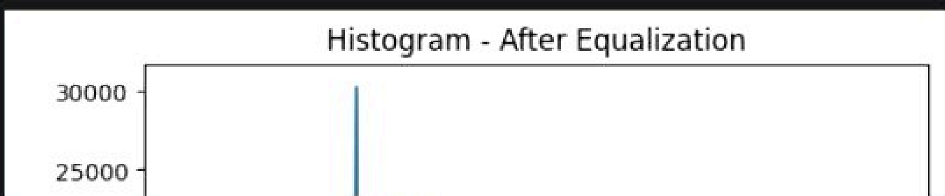


[17] 0s

```
# ============================================
# Histogram of equalized image
# ============================================

hist_eq = cv2.calcHist([equalized], [0], None, [256], [0,256])

plt.figure(figsize=(6,4))
plt.plot(hist_eq)
plt.title("Histogram - After Equalization")
plt.xlabel("Intensity")
plt.ylabel("Pixel Count")
plt.show()
```



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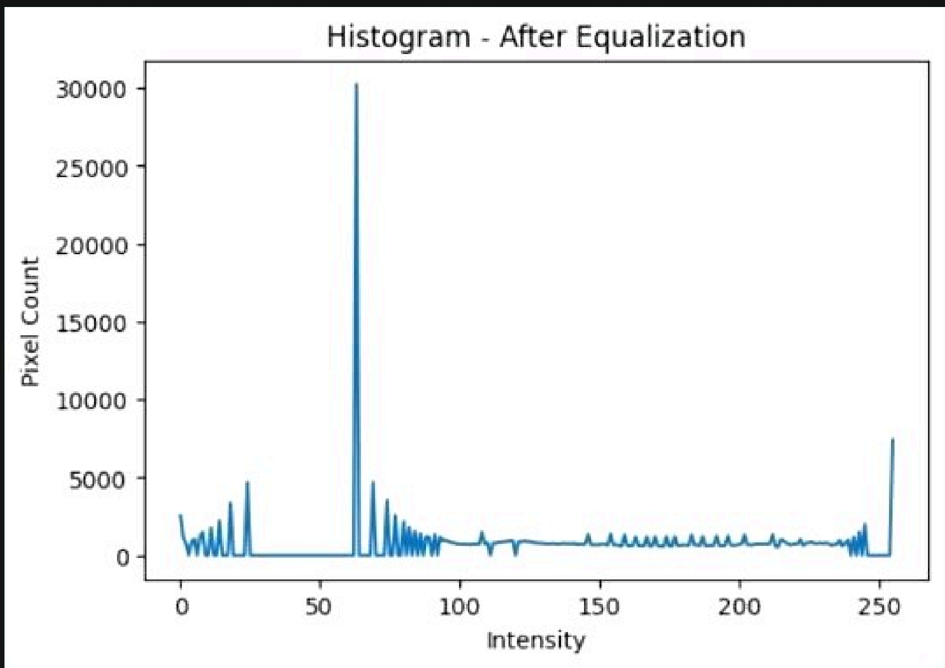
[17] 0s

```python
# ==========================================
# Histogram of equalized image
# ==========================================

hist_eq = cv2.calcHist([equalized], [0], None, [256], [0,256])

plt.figure(figsize=(6,4))
plt.plot(hist_eq)
plt.title("Histogram - After Equalization")
plt.xlabel("Intensity")
plt.ylabel("Pixel Count")
plt.show()
```



[18] 0s

```python
# ==========================================
# Showing sample pixel values (data)
# ==========================================

print("===== Sample grayscale values (first 5x5 pixels) =====")
print(gray[:5, :5])

print("\n===== Sample equalized values (first 5x5 pixels) =====")
print(equalized[:5, :5])
```

```
===== Sample grayscale values (first 5x5 pixels) =====
[[23 23 23 23 23]
 [12 12 12 12 12]
 [10 10 10 10 10]
 [13 13 13 13 13]
 [11 11 11 11 11]]

===== Sample equalized values (first 5x5 pixels) =====
[[93 93 93 93 93]
```

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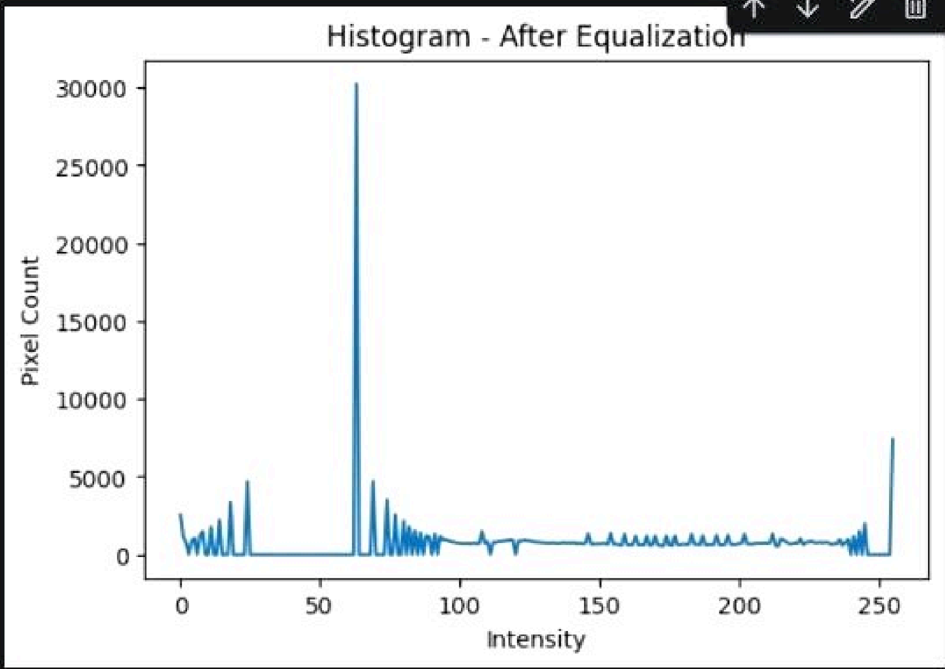

[18] 0s

```
# ==========================================
# Showing sample pixel values (data)
# ==========================================

print("===== Sample grayscale values (first 5x5 pixels) =====")
print(gray[:5, :5])

print("\n===== Sample equalized values (first 5x5 pixels) =====")
print(equalized[:5, :5])
```

```
===== Sample grayscale values (first 5x5 pixels) =====
[[23 23 23 23 23]
 [12 12 12 12 12]
 [10 10 10 10 10]
 [13 13 13 13 13]
 [11 11 11 11 11]]

===== Sample equalized values (first 5x5 pixels) =====
[[93 93 93 93 93]
 [63 63 63 63 63]
 [18 18 18 18 18]
 [69 69 69 69 69]
 [24 24 24 24 24]]
```

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[4] 1m

```
# --- Import libraries ---
import cv2
from google.colab.patches import cv2_imshow
import numpy as np

# --- Upload image in Colab ---
from google.colab import files
uploaded = files.upload()  # پنجره آپلود باز میشه و تصویر رو انتخاب می‌کنید

# Get the uploaded file name
filename = list(uploaded.keys())[0]

# --- Load image ---
img = cv2.imread(filename)
if img is None:
    raise ValueError("Image not found or invalid!")

# All subsequent image processing should not be indented under the if-s
# --- 1. Show original image ---
cv2_imshow(img)

# --- 2. Gaussian Blur ---
gaussian_blur = cv2.GaussianBlur(img, (15, 15), 0)
cv2_imshow(gaussian_blur)

# --- 3. Average Blur ---
average_blur = cv2.blur(img, (10, 10))
cv2_imshow(average_blur)

# --- 4. Edge Detection (Canny) ---
edges = cv2.Canny(img, 100, 200)
cv2_imshow(edges)

# --- 5. Convert to Grayscale ---
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2_imshow(gray)

# --- 6. Sharpening Filter ---
kernel_sharpen = np.array([[0,-1,0],
                           [-1,5,-1],
                           [0,-1,0]])
sharpened = cv2.filter2D(img, -1, kernel_sharpen)
cv2_imshow(sharpened)

# --- 7. Optional: Sepia Effect ---
sepia_filter = np.array([[0.272, 0.534, 0.131],
                         [0.349, 0.686, 0.168],
                         [0.393, 0.769, 0.189]])
sepia = cv2.transform(img, sepia_filter)
sepia = np.clip(sepia, 0, 255).astype(np.uint8)
cv2_imshow(sepia)
```

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[4] 1m

```
# --- 5. Convert to Grayscale ---
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2_imshow(gray)

# --- 6. Sharpening Filter ---
kernel_sharpen = np.array([[0,-1,0],
                           [-1,5,-1],
                           [0,-1,0]])
sharpened = cv2.filter2D(img, -1, kernel_sharpen)
cv2_imshow(sharpened)

# --- 7. Optional: Sepia Effect ---
sepia_filter = np.array([[0.272, 0.534, 0.131],
                         [0.349, 0.686, 0.168],
                         [0.393, 0.769, 0.189]])
sepia = cv2.transform(img, sepia_filter)
sepia = np.clip(sepia, 0, 255).astype(np.uint8)
cv2_imshow(sepia)

cv2.destroyAllWindows()
```

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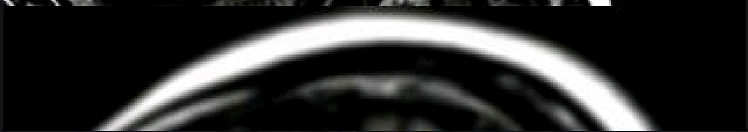
Saving IMG_20260105_234459_134.jpg to IMG_20260105_234459_134.jpg



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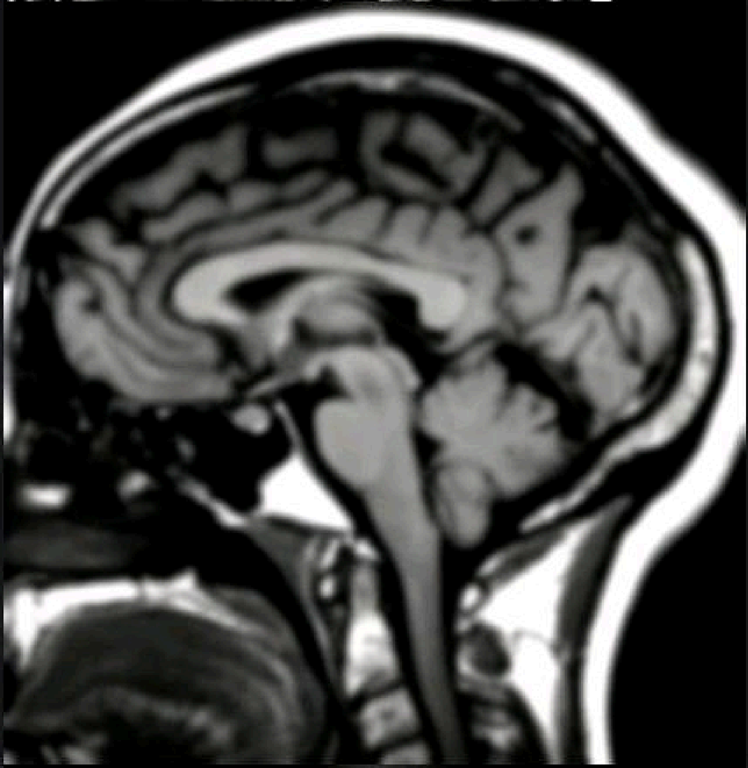
```
Saving IMG_20260105_234459_134.jpg to IMG_20260105_234459_134.jpg
```



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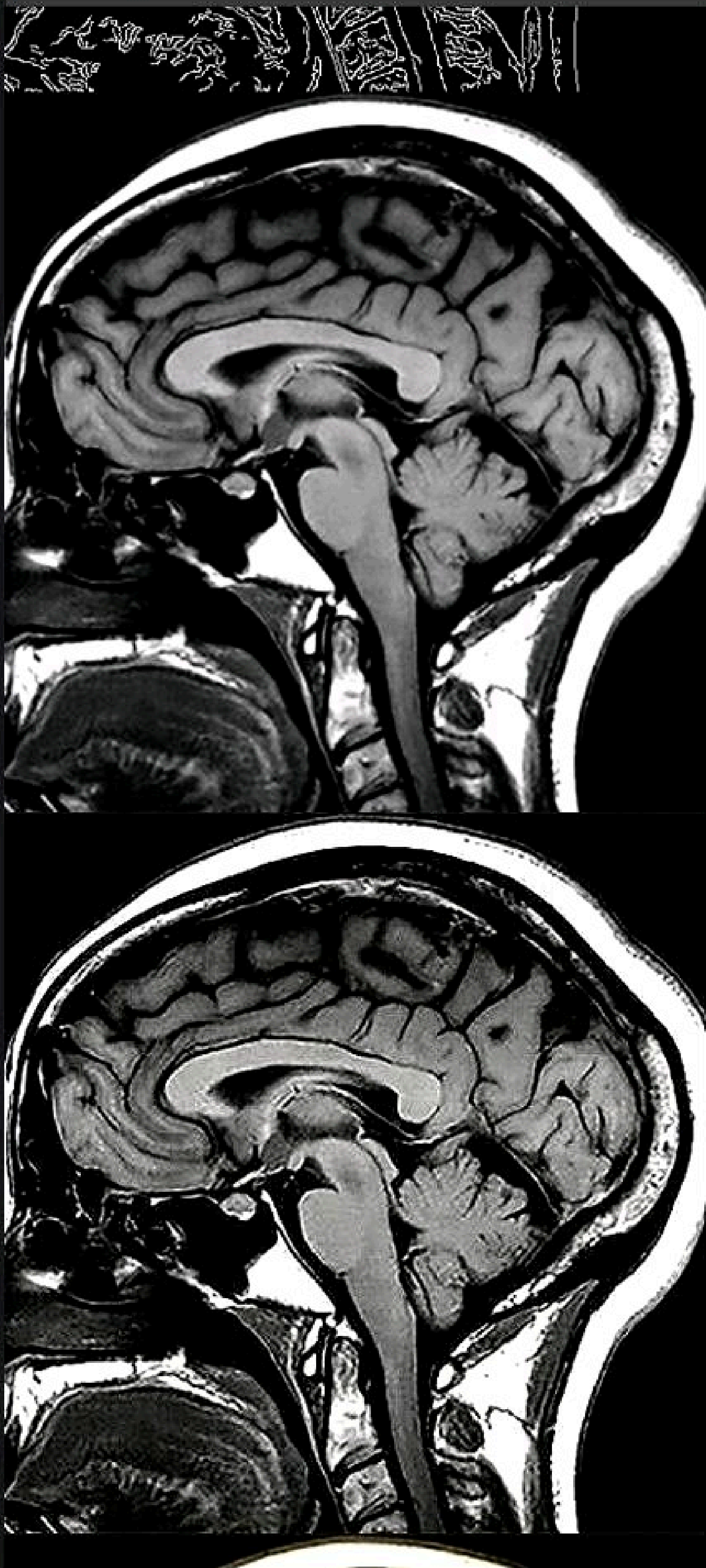
1m



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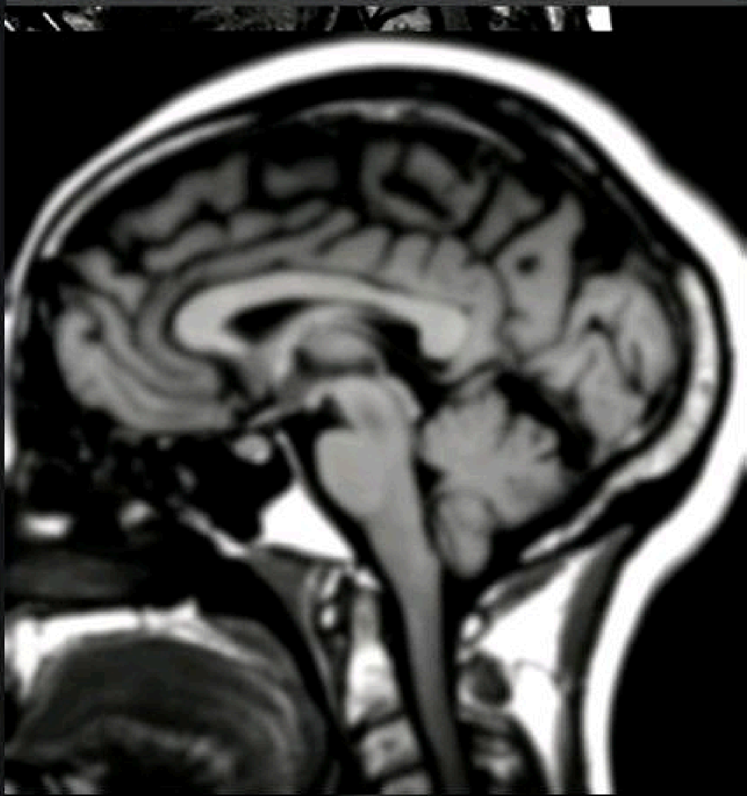
[4]
1m

```
cv2_imshow(sepia)

cv2.destroyAllWindows()
```



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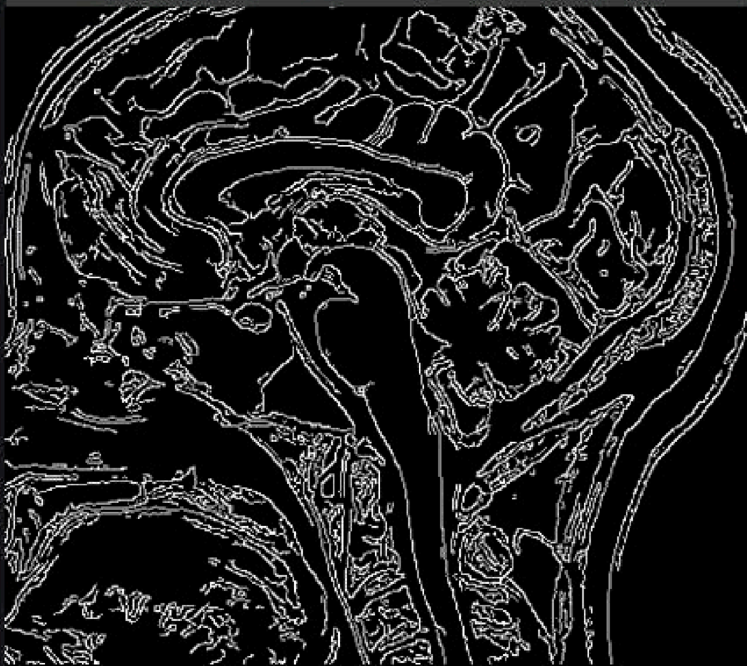
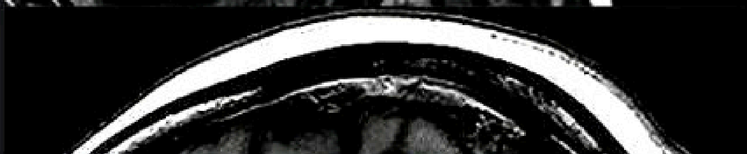
[4] 1m

```
cv2_imshow(sepia)

cv2.destroyAllWindows()
```





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